



Mechatronics and Robotics Engineering Technology

ROBT 4456: PLC Applications

PROJECT REPORT: ADVANCED FOUR-FLOOR ELEVATOR

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Abstract

This PLC project demonstrates the application of various PLC programming techniques through the implementation of a four-floor elevator controller. The project programs the lab bench mock elevator to behave as a real elevator using a Method 3 finite state machine to manage movement, door control, emergency stops, and multi-floor call queuing. Additional unique features were implemented including two Add-On Instructions (FlashingLight and FloorStats), a User-Defined Type for structured fault logging, a dedicated periodic fault detection program, and a bonus HMI Penthouse routine implementing PIN-gated access to Floor 4. All nine project parts are complete and documented.

Preface

The assignment of this project was provided in 9 distinct parts. Each part asks for additional functionality and must be completed sequentially. All 9 parts are complete and documented. Parts 1 through 6 establish core elevator operation. Part 7 adds statistical analysis via the FloorStats AOI. Part 8 implements fault detection in a separate periodic task with UDT-based logging. Part 9 implements the HMI Penthouse bonus feature.

Contents

Definitions

Table o.1: List of Abbreviations

Abbreviation	Definition
PLC	Programmable Logic Controller
FSM	Finite State Machine
AOI	Add-On Instruction
UDT	User-Defined Type
HMI	Human-Machine Interface
MCR	Master Control Reset
TON	Timer On-Delay
ONS	One-Shot
GSV	Get System Value
JSR	Jump to Subroutine

Table o.2: List of Symbols

Symbol	Definition
V:	Virtual – prefix for buffered I/O signal tags
!	Boolean operator: Logical NOT
.DN	Timer or Counter Done bit
.ACC	Accumulated value register
.TT	Timer Timing bit

1 Introduction

This document discusses the design and implementation of an Advanced Four-Floor Elevator PLC controller. As a PLC programming project, the hardware is provided by BCIT Mechatronics and Robotics.

1.1 Background

Elevator control systems are a practical application of discrete-event automation. A PLC is well-suited to this task due to its deterministic scan-cycle, reliable digital I/O handling, and support for structured programming. This project applies those principles to a lab bench mock elevator with four floors, a motorized door, floor call buttons, an inside request panel, and an emergency stop circuit.

1.2 Project Description

The elevator project is defined in nine parts. Each part demands additional features on top of the previous parts. This report reflects the most advanced implementation across all nine parts.

1.3 Project Hardware

The hardware design and implementation is provided at the start of the project. The main PLC controller is an Allen-Bradley 1769 CompactLogix PLC. The I/O modules utilized are:

Table 1.1: I/O Module Summary

Module	Type	Function
Local:2	AB 1769-DI16 – Digital Input	Pushbuttons (PB1–PB4, PBO, PBC, PBE5), floor pushbuttons (FPB1–FPB4), START, STOP
Local:3	AB 1769-DI16 – Digital Input	Floor limit switches (FLS1–FLS4), door limit switches (DCLS, DOLS), toggle switches (TS0–TS2)
Local:4	AB 1769-DO16 – Digital Output	Indicator lights (IL1–IL4, ILO, ILC, ILE5), floor indicators (FIL1–FIL4), fault indicators (ILTS0–ILTS2)
Local:5	AB 1769-DO16 – Digital Output	Motor 0 (V_M0), Control Relay 0 (V_CR0), Solenoid 0 (V_SOL0)

2 Project Overview

This section discusses the organization of the project.

2.1 Studio 5000 Project Organization

The PLC program is organized to be modular. The Studio 5000 project (ClassicElevator) is organized into two tasks:

- Continuous Task (MainTask): Contains MainProgram, which holds all elevator control routines called sequentially from MainRoutine.
- Periodic Task (100 ms): Contains FaultProgram, which holds the fault condition monitoring routine, executing independently of elevator scan time.

The MainRoutine is the master routine within MainProgram. Emergency-related subroutines (StateMachine_Pre, EmergencyStop, and the emergency indicator rungs) are placed outside the MCR block to ensure they always execute regardless of emergency state.

2.2 Controller Tags

Table 2.1: Controller Tags

Tag	Type	Description
Local:2:I.Data	DINT	Module 2 physical input register (buffered to V_2I_Data)
Local:3:I.Data	DINT	Module 3 physical input register (buffered to V_3I_Data)
Local:4:O.Data	DINT	Module 4 physical output register (driven by V_4O_Data)
Local:5:O.Data	DINT	Module 5 physical output register (driven by V_5O_Data)
Current_Floor_Idle	DINT	Floor the elevator is currently at or last stopped at
HMI_Pin_Input	DINT	PIN value entered via HMI numeric input field
HMI_Pin_Enter	BOOL	Momentary bit driven by HMI submit button
HMI_Pin_Required	BOOL	Set when Floor 4 is requested without a valid PIN
HMI_Pin_Correct	BOOL	Set when the entered PIN matches the correct value
HMI_Access	BOOL	General HMI access flag

2.3 Ladder Logic Structure

Each program adheres to an Input Buffer, Logic, and Output Buffer structure. The input buffer routine captures physical I/O into virtual V_ tags at the start of the scan, ensuring

logical inputs do not change during a single scan. The output buffer only updates after the rest of the program has been scanned. Each subroutine is responsible for a dedicated set of tags which may not be written to in other subroutines.

2.4 Custom Defined Structures

This project uses two Add-On Instructions: FlashingLight and FloorStats. This project uses one User-Defined Type: FaultLog. These are documented in detail in Sections 4 and 5 respectively.

3 Main Program

The elevator operation is managed by a Method 3 state machine. This state machine determines logical movement of the elevator by driving the motor and its direction. The Main Program (MainProgram) runs in the continuous task and contains all elevator control routines.

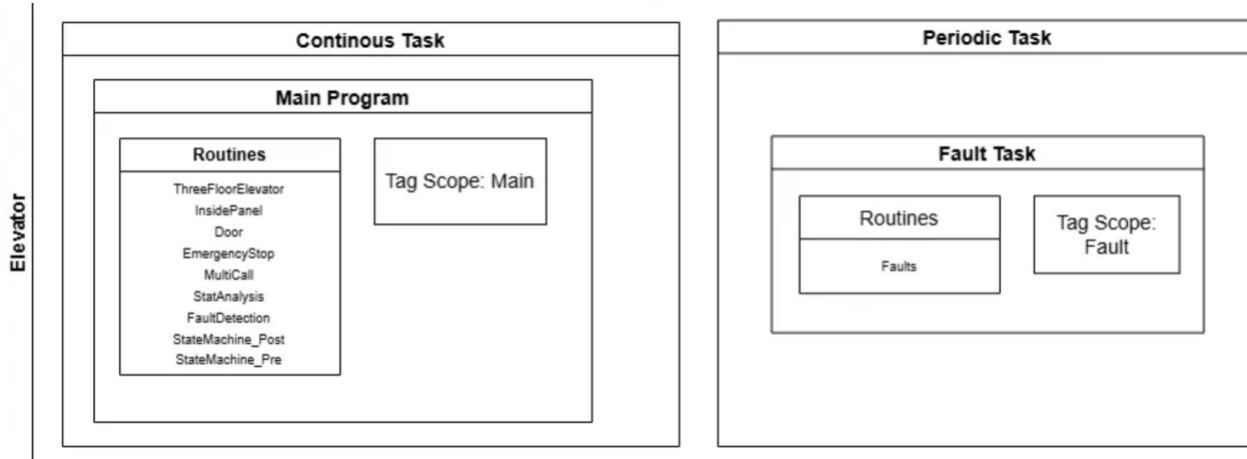


Figure2: Program flow

3.1 Program Tags

These tags are internal to the Main Program. Access to physical I/O tags is managed by the input and output buffer subroutines.

Table 3.1: Controller Tag Buffering

Tag	Mapping	Type	Subroutine
V_2I_Data	Local:2:I.Data	DINT	InputBuffer
V_3I_Data	Local:3:I.Data	DINT	InputBuffer
V_4O_Data	Local:4:O.Data	DINT	OutputBuffer
V_5O_Data	Local:5:O.Data	DINT	OutputBuffer

Table 3.2: Input Module 2 – InputBuffer Tags

Tag	Alias	Type	Description
V_PB4	V_2I_Data.0	BOOL	V: Pushbutton 4 (in-car Floor 4 request)
V_PB3	V_2I_Data.1	BOOL	V: Pushbutton 3
V_PB2	V_2I_Data.2	BOOL	V: Pushbutton 2
V_PB1	V_2I_Data.3	BOOL	V: Pushbutton 1

Tag	Alias	Type	Description
V_PBO	V_2I_Data.4	BOOL	V: Pushbutton Open
V_PBC	V_2I_Data.5	BOOL	V: Pushbutton Close
V_PBE5	V_2I_Data.6	BOOL	V: Pushbutton Emergency 5
V_FPB4	V_2I_Data.8	BOOL	V: Floor Pushbutton 4
V_FPB3	V_2I_Data.9	BOOL	V: Floor Pushbutton 3
V_FPB2	V_2I_Data.10	BOOL	V: Floor Pushbutton 2
V_FPB1	V_2I_Data.11	BOOL	V: Floor Pushbutton 1
V_START	V_2I_Data.14	BOOL	V: START (N.O.) – emergency reset
V_STOP	V_2I_Data.15	BOOL	V: STOP (N.C.)

Table 3.3: Input Module 3 – InputBuffer Tags

Tag	Alias	Type	Description
V_FLS1	V_3I_Data.0	BOOL	V: Floor Limit Switch 1
V_FLS2	V_3I_Data.1	BOOL	V: Floor Limit Switch 2
V_FLS3	V_3I_Data.2	BOOL	V: Floor Limit Switch 3
V_FLS4	V_3I_Data.3	BOOL	V: Floor Limit Switch 4
V_DCLS	V_3I_Data.8	BOOL	V: Door Close Limit Switch
V_DOLS	V_3I_Data.9	BOOL	V: Door Open Limit Switch
V_TS2	V_3I_Data.10	BOOL	V: Toggle Switch 2
V_TS1	V_3I_Data.11	BOOL	V: Toggle Switch 1
V_TS0	V_3I_Data.12	BOOL	V: Toggle Switch 0

Table 3.4: Output Module 4 – OutputBuffer Tags

Tag	Alias	Type	Description
V_IL4	V_4O_Data.0	BOOL	V: Indicator Light 4
V_IL3	V_4O_Data.1	BOOL	V: Indicator Light 3
V_IL2	V_4O_Data.2	BOOL	V: Indicator Light 2
V_IL1	V_4O_Data.3	BOOL	V: Indicator Light 1
V_ILO	V_4O_Data.4	BOOL	V: Indicator Light Open
V_ILC	V_4O_Data.5	BOOL	V: Indicator Light Close
V_ILE5	V_4O_Data.6	BOOL	V: Indicator Light Emergency 5

Tag	Alias	Type	Description
V_FIL4	V_4O_Data.8	BOOL	V: Floor Indicator Light 4
V_FIL3	V_4O_Data.9	BOOL	V: Floor Indicator Light 3
V_FIL2	V_4O_Data.10	BOOL	V: Floor Indicator Light 2
V_FIL1	V_4O_Data.11	BOOL	V: Floor Indicator Light 1
V_ILTS0	V_4O_Data.14	BOOL	V: Indicator Light Toggle Switch 0 (DoorFault)
V_ILTS1	V_4O_Data.13	BOOL	V: Indicator Light Toggle Switch 1 (MotorFault)
V_ILTS2	V_4O_Data.12	BOOL	V: Indicator Light Toggle Switch 2 (PendReqFault)

Table 3.5: Output Module 5 – OutputBuffer Tags

Tag	Alias	Type	Description
V_M0	V_5O_Data.0	BOOL	V: Motor 0 (run elevator motor)
V_SOL0	V_5O_Data.2	BOOL	V: Solenoid 0 (door actuator)
V_CR0	V_5O_Data.4	BOOL	V: Control Relay 0 (direction: 1 = Up)

Table 3.6: State Descriptions

State	Description	Entry Conditions	Exit Conditions	Actions
1 – Idle	Elevator stationary, waiting for a call	First scan; State 3 or 5 on arrival	Call queued in Request[]	Motor off; indicators off
2 – Move Up	Elevator travelling upward to Next_Floor	State 1/3: Next_Floor > Current_Floor_Idle	V_FLS[Next_Floor] active	V_M0 = 1; V_CR0 = 1
3 – Move Down	Elevator travelling downward to Next_Floor	State 1/2: Next_Floor < Current_Floor_Idle	V_FLS[Next_Floor] active	V_M0 = 1; V_CR0 = 0
4 – Door Open	Door open; passengers boarding or exiting	Arrival at Next_Floor	TimerA.DN and close conditions met	V_SOL0 = 1; V_ILO active
5 – Door Closing	Door closing after service	State 4: TimerA.DN	V_DCLS active	V_ILC active

State	Description	Entry Conditions	Exit Conditions	Actions
100 – Emergency	Emergency stop; elevator between floors	V_PBE5 pressed while moving	V_START pressed (no major fault)	All motion stopped; V_ILE5 flashes
400 – Emerg Floor	Emergency stop; elevator at a floor	V_PBE5 pressed while at floor limit switch	V_START pressed (no major fault)	Door held open; V_ILE5 flashes

3.2 MainRoutine

MainRoutine is the master routine that calls all subroutines in sequence each scan. On the first scan (s:fs), it initializes Current_State = 1 (Idle), Next_State = 1, Next_Floor = Floor_1, and immediately sets Next_State = State3_MoveDown so the elevator homes to Floor 1. Each subsequent scan, subroutines are called in the following order:

Table 3.7: Subroutine Call Order in MainRoutine

Order	Subroutine	MCR Position	Description
1	FaultDetection	Before MCR	Reads FaultProgram flags; logs UDT entries; drives fault indicator lights
2	StateMachine_Pre	Before MCR	Emergency transitions; always executes regardless of MCR state
—	(MCR open)	—	Normal subroutines execute only when not in emergency state
3	StatAnalysis	Inside MCR	Per-floor call counts, service times, timestamps; calls FloorStats AOIs
4	HMI_PentHouse	Inside MCR	PIN-gated Floor 4 access; manages RequestHMI[4] and FlashFloor4
5	MultiCall	Inside MCR	Accepts all floor calls and requests; sets Request[], Move_Up, Move_Down
6	InsidePanel	Inside MCR	Drives in-car indicator lights IL1–IL3 based on Request[] state
7	Door	Inside MCR	Door solenoid, TimerA/B, ILO/ILC indicators, PBO hold-open logic
8	ThreeFloorElevator	Inside MCR	Resolves Next_Floor; tracks Current_Floor_Idle; drives V_M0 and V_CR0

Order	Subroutine	MCR Position	Description
9	StateMachine_Post	Inside MCR	Evaluates floor arrival and door transitions; updates Next_State
—	(MCR close)	—	End of normal operation block
10	EmergencyStop	After MCR	In State 400: calls Door to keep door open during emergency at floor
11	Rungs 15–17	After MCR	Drives FlashingLight AOI and V_ILE5 in any emergency state

3.3 ELEVATORSTATEMACHINE

This subroutine is responsible for the main elevator operation. The elevator actuators are only controlled by the logic within this subroutine and is also the only subroutine which may call the door state machine subroutine. The logic within this subroutine implements a finite state machine.

Table 0.1: ElevatorStateMachine Tags

Tag	Alias	Description	Type	Read By
Current_State	Base	Current State of the FSM	DINT	
Next_State	Base	Accepted Next State of the FSM	DINT	
V_M0	V_5O_Data.0	V: Motor 0	BOOL	OutputBuffer
V_CR0	V_5O_Data.2	V: Control Relay 0	BOOL	OutputBuffer

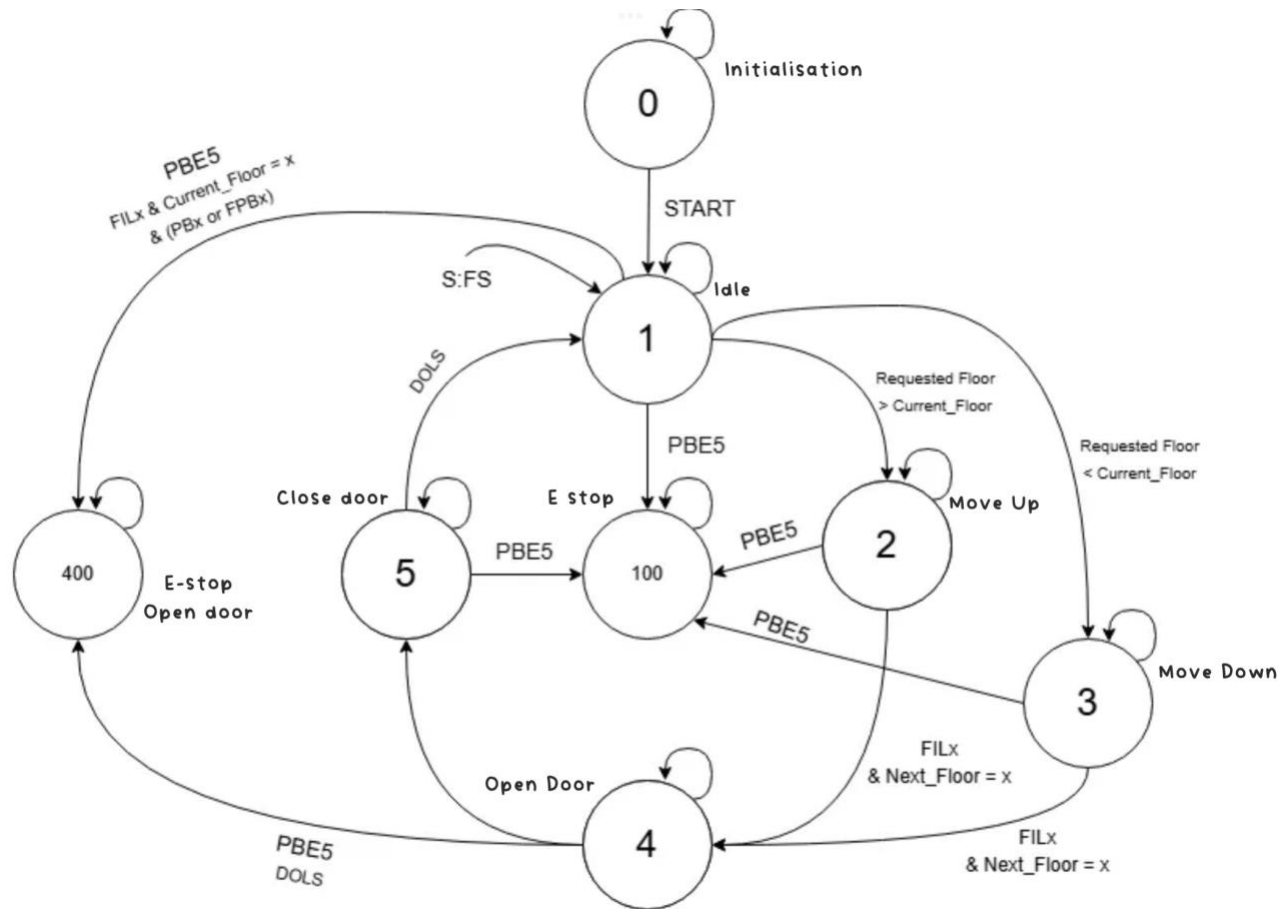


Figure 0 A State Machine

The state machine forces the elevator to behave a specific way. Modifying the transitions can modify the overall behaviour of the system.

3.3 Call and Request Manager

Subroutine: MultiCall. Passengers press buttons at a floor to call the elevator to that floor. Once inside, the passenger may request a floor. This subroutine accepts all calls and requests and determines the next floor the elevator will travel to.

- Implementation: Uses GRT and LES instructions to compare active requests against Current_Floor_Idle. RequestUp[N] and RequestDown[N] are derived per floor and passed to ThreeFloorElevator for Next_Floor resolution. Calls are blocked in emergency states.

Table 3.8: MultiCall Tags

Tag	Alias / Mapping	Type	Description
Current_State	—	DINT	Read: calls blocked in emergency states 100 and 400
Next_State	—	DINT	Read: dual-state emergency check
Current_Floor_Idle	—	DINT	Read: calls blocked when elevator already at that floor
Request[1..4]	—	BOOL[32]	Written: sealed-in call/request flags per floor
RequestHMI[4]	—	BOOL[32]	Read: HMI-originated Floor 4 request (from HMI_PentHouse)
RequestUp[2..4]	—	BOOL[32]	Written: request flags for floors above current
RequestDown[1..3]	—	BOOL[32]	Written: request flags for floors below current
Move_Up	—	BOOL	Written: set when any request exists above current floor
Move_Down	—	BOOL	Written: set when any request exists below current floor
V_FPB1	V_2I_Data.11	BOOL	Floor Pushbutton 1 – external call Floor 1
V_FPB2	V_2I_Data.10	BOOL	Floor Pushbutton 2
V_FPB3	V_2I_Data.9	BOOL	Floor Pushbutton 3
V_FPB4	V_2I_Data.8	BOOL	Floor Pushbutton 4
V_PB1	V_2I_Data.3	BOOL	In-car Pushbutton 1 – internal request Floor 1
V_PB2	V_2I_Data.2	BOOL	In-car Pushbutton 2
V_PB3	V_2I_Data.1	BOOL	In-car Pushbutton 3
V_FIL1	V_4O_Data.11	BOOL	Written: Floor Indicator Light 1
V_FIL2	V_4O_Data.10	BOOL	Written: Floor Indicator Light 2
V_FIL3	V_4O_Data.9	BOOL	Written: Floor Indicator Light 3

3.4 Door Routine

Subroutine: Door. This routine manages the door controls and opens and closes while the elevator is servicing floors.

- Implementation: TimerA (5 s) runs in State 4; TimerA.DN signals the door-close transition. V_PBO sets PBO_ON and resets TimerB to extend the open period by 3

s. After 2 s, V_PBC can initiate early close. V_SOLO energized in State 4 and State 400. The /TimerA.DN interlock was intentionally removed from the TON rung to prevent TimerA from resetting on its own Done bit.

Table 3.9: Door Tags

Tag	Alias / Mapping	Type	Description
Current_State	—	DINT	Read: door logic runs in State 4 and 400
Next_State	—	DINT	Read: dual-state check
TimerA	—	TIMER	5 s door open timer; .DN used by StateMachine_Post for close transition
TimerB	—	TIMER	3 s PBO hold-open extension timer; runs when PBO_ON set and TimerA.DN
PBO_ON	—	DINT	Read/Written: set when V_PBO pressed in State 4; cleared on door close
PBC_ON	—	DINT	Read/Written: set when V_PBC pressed after 2 s in State 4
DoorOpen	—	DINT	Read/Written: high when V_DOLS is active
V_PBO	V_2I_Data.4	BOOL	Pushbutton Open – sets PBO_ON and resets TimerB
V_PBC	V_2I_Data.5	BOOL	Pushbutton Close – sets PBC_ON after 2 s
V_DOLS	V_3I_Data.9	BOOL	Door Open Limit Switch
V_DCLS	V_3I_Data.8	BOOL	Door Close Limit Switch
V_SOLO	V_5O_Data.2	BOOL	Written: door solenoid – energized in State 4 and 400
V_ILO	V_4O_Data.4	BOOL	Written: Indicator Light Open – on when V_DOLS active and V_PBO not pressed
V_ILC	V_4O_Data.5	BOOL	Written: Indicator Light Close – on when V_DCLS active and V_PBC not pressed

3.5 Emergency Stop

Subroutine: EmergencyStop. This routine manages the emergency stop condition and keeps the door open when the elevator is at a floor during an emergency.

- Implementation: StateMachine_Pre forces Next_State = 100 when V_PBE5 is pressed while moving, or Next_State = 400 when pressed at a floor. EmergencyStop (called after MCR) calls the Door subroutine in State 400 to hold

the door open. The FlashingLight AOI drives V_ILE5 at 700 ms in either emergency state.

Table 3.10: EmergencyStop Tags

Tag	Alias / Mapping	Type	Description
Current_State	—	DINT	Read: checked for State 400
Next_State	—	DINT	Read: dual-state check for State 400
State400_DorOpenEmerg	—	DINT	Read: constant = 400 used in EQU comparison
V_PBE5	V_2I_Data.6	BOOL	Virtual Pushbutton Emergency 5 – evaluated in StateMachine_Pre
V_START	V_2I_Data.14	BOOL	Virtual START button – clears emergency state in StateMachine_Pre
Flash	—	BOOL	Written by FlashingLight AOI; drives V_ILE5 in emergency states
V_ILE5	V_4O_Data.6	BOOL	Written: Indicator Light Emergency 5 – flashes at 700 ms

4 Implementation: Main Program Subroutines

This section describes each subroutine by project part, including the relevant tags and implementation approach.

4.1 Part 1: Three-Floor Elevator

Subroutine: ThreeFloorElevator. Controls basic elevator response to calls from floors 1–3. Ensures the elevator moves to Floor 1 upon initial run and never travels outside the 1–3 floor range.

- Implementation: Uses EQU and MOV instructions to compare the requested floor against Current_Floor_Idle and update Next_Floor. Priority rungs resolve the nearest floor in the direction of travel.

Table 4.1: ThreeFloorElevator Tags

Tag	Alias / Mapping	Type	Description
Current_State	—	DINT	Read: motor/direction only output in States 2 and 3
Next_State	—	DINT	Read: dual-state check
Next_Floor	—	DINT	Written: target floor resolved from request priority logic
Current_Floor_Idle	—	DINT	Written: updated when floor limit switch activates during movement
Move_Up	—	BOOL	Read: direction flag from MultiCall
Move_Down	—	BOOL	Read: direction flag from MultiCall
Request[1..4]	—	BOOL[32]	Read: active call/request flags per floor
RequestUp[2..4]	—	BOOL[32]	Read: upward requests used in priority resolution
RequestDown[1..3]	—	BOOL[32]	Read: downward requests used in priority resolution
V_FLS1	V_3I_Data.0	BOOL	Floor 1 limit switch – updates Current_Floor_Idle and triggers arrival
V_FLS2	V_3I_Data.1	BOOL	Floor 2 limit switch
V_FLS3	V_3I_Data.2	BOOL	Floor 3 limit switch
V_FLS4	V_3I_Data.3	BOOL	Floor 4 limit switch
V_M0	V_5O_Data.0	BOOL	Written: motor run output (on in States 2 and 3)

Tag	Alias / Mapping	Type	Description
V_CR0	V_5O_Data.4	BOOL	Written: direction relay (1 = Up, 0 = Down)

4.2 Part 2: Inside Panel

Subroutine: InsidePanel. Processes passenger requests made from within the elevator car for floors 1–3.

- Implementation: Sealed-in latch logic latches internal requests into the Request array alongside external floor calls. The indicator light stays on until Request[N] clears when the floor is serviced.

Table 4.2: InsidePanel Tags

Tag	Alias / Mapping	Type	Description
Request[1]	—	BOOL[32]	Read/seal: Floor 1 request active flag
Request[2]	—	BOOL[32]	Read/seal: Floor 2 request active flag
Request[3]	—	BOOL[32]	Read/seal: Floor 3 request active flag
V_PB1	V_2I_Data.3	BOOL	In-car Pushbutton 1 – sets indicator and request
V_PB2	V_2I_Data.2	BOOL	In-car Pushbutton 2
V_PB3	V_2I_Data.1	BOOL	In-car Pushbutton 3
V_IL1	V_4O_Data.3	BOOL	Written: Indicator Light 1
V_IL2	V_4O_Data.2	BOOL	Written: Indicator Light 2
V_IL3	V_4O_Data.1	BOOL	Written: Indicator Light 3

4.3 Part 3: Elevator Door

Subroutine: Door. Manages the 5-second open/close cycle when the elevator arrives at a floor. See Table 3.9 for the full tag listing.

- Implementation: TimerA (5 s) provides the default open duration. TimerB (3 s) extends the door-open period if V_PBO is pressed. After 2 s, V_PBC can close the door early. The /TimerA.DN interlock was removed from the TON rung to prevent self-reset on Done.

4.4 Part 4: Emergency Stop

Subroutine: EmergencyStop. Halts all operation upon pressing V_PBE5. Prevents the elevator from responding to any new calls until START is pressed to reset the system. See Table 3.10 for the full tag listing.

- Implementation: StateMachine_Pre immediately moves the FSM to State100_Emerg (moving) or State400_DorOpenEmerg (at floor). Both StateMachine_Pre and EmergencyStop are placed outside the MCR block so emergency transitions always fire.

4.5 Part 5: Four-Floor Elevator

Subroutine: ThreeFloorElevator and StateMachine_Post (extended). Extends the system range to include Floor 4 for both calls and requests.

- Tags: V_FPB4, V_PB4, V_FIL4, V_IL4, V_FLS4.
- Implementation: Limit switch comparisons updated to include Floor 4. Request array extended to index 4. Next_Floor resolution logic in ThreeFloorElevator updated with additional priority rungs for Floor 4.

4.6 Part 6: Multiple Calls/Requests

Subroutine: MultiCall. Manages simultaneous requests by prioritizing floors in the current direction of travel until the highest or lowest request is serviced. See Table 3.8 for the full tag listing.

- Implementation: GRT and LES instructions compare active requests against Current_Floor_Idle to determine travel direction. The elevator maintains direction until the outermost request in that direction is reached, then reverses.

4.7 Part 7: Statistical Analysis

Subroutine: StatAnalysis. Monitors usage metrics including the number of calls per floor and average service time.

- Implementation: Employs the FloorStats Add-On Instruction (one instance per floor: FloorStats1–FloorStats4). A Service_Timer starts at the call and ends when the door closes at the destination. A GSV WallClockTime instruction records the timestamp into LastServiced[[]].

Table 4.3: StatAnalysis Tags

Tag	Alias / Mapping	Type	Description
Next_Floor	—	DINT	Read: determines which floor AOI to call
NumCall[1..4]	—	DINT[5]	Written by AOI: call count per floor

Tag	Alias / Mapping	Type	Description
AvgServTime[1..4]	—	DINT[5]	Written by AOI: average service time per floor (ms)
TotalTime[1..4]	—	DINT[5]	Written by AOI: accumulated service time per floor (ms)
ServiceTime[1..4]	—	DINT[5]	Written by AOI: most recent service time per floor (ms)
ServiceTimer[1..4]	—	TIMER[5]	AOI internal: service duration timers (preset 30 000 ms)
CallBit[1..4]	—	DINT[5]	Written: first-call trigger bit; set by ONS, cleared by AOI
DoneBit[1..4]	—	BOOL[32]	Written: door-closed confirmation latch per floor
DoorClosed[1..4]	—	BOOL[32]	Written: door-closed state latch per floor
LastServed[N,0..6]	—	DINT[5,7]	Written: GSV wall clock timestamp of last service per floor
FloorStats1	—	FloorStats	AOI instance – floor 1 statistics
FloorStats2	—	FloorStats	AOI instance – floor 2 statistics
FloorStats3	—	FloorStats	AOI instance – floor 3 statistics
FloorStats4	—	FloorStats	AOI instance – floor 4 statistics
V_DCLS	V_3I_Data.8	BOOL	Door Close Limit Switch – triggers DoorClosed latch
V_DOLS	V_3I_Data.9	BOOL	Door Open Limit Switch – sets DoorOpen flag
V_FPB1	V_2I_Data.11	BOOL	Floor Pushbutton 1 – call detection for CallBit
V_FPB2	V_2I_Data.10	BOOL	Floor Pushbutton 2
V_FPB3	V_2I_Data.9	BOOL	Floor Pushbutton 3
V_FPB4	V_2I_Data.8	BOOL	Floor Pushbutton 4
V_PB1	V_2I_Data.3	BOOL	In-car Pushbutton 1
V_PB2	V_2I_Data.2	BOOL	In-car Pushbutton 2
V_PB3	V_2I_Data.1	BOOL	In-car Pushbutton 3
V_PB4	V_2I_Data.0	BOOL	In-car Pushbutton 4

Table 4.4: FloorStats AOI Parameters

Parameter	Usage	Type	Description
Call_Bit	InOut	DINT	Set to 1 on first button press (ONS); cleared by AOI after counting
Num_Call	InOut	DINT	Running count of calls to this floor
Door_Closed	InOut	BOOL	Enable signal; high when door has just closed after servicing this floor
Done_Bit	InOut	BOOL	Latch to prevent repeated AOI execution per service event
Service_Time	InOut	DINT	Service_Timer.ACC captured at door close
Total_Time	InOut	DINT	Accumulated service time across all calls
Avg_ServTime	InOut	DINT	Total_Time / Num_Call; equals Total_Time if Num_Call = 0
Service_Timer	Local	TIMER	Internal TON (30 s preset) timing from call to door close

5 Program A: Fault Program

Program A (FaultProgram) executes within a Periodic task every 100 ms. This program detects faults based on monitored system conditions and logs them using a User-Defined Type. The periodic task ensures fault monitoring is independent of the continuous elevator scan.

5.1 Program Tags

Table 5.1: FaultProgram Tags

Tag	Data Type	Description
Faults[0..3]	FaultTypes[4]	Struct of fault condition BOOLs: MotorOnFault, DoorFault, PendReqFault, SensorFault, MovEmergFault
MajorFault	BOOL	High when a major fault is active; blocks emergency reset in StateMachine_Pre
MotorFaultTimer	TIMER	TON – motor run duration monitor (preset 10 000 ms)
DoorFaultTimer	TIMER	TON – door open duration monitor (preset 20 000 ms)
TimerReq[1..4]	TIMER[5]	TON per floor – request age timers (preset 15 000 ms each)

5.2 Part 8: Fault Detection

Subroutine: FaultProgram (Periodic Task, 100 ms). A dedicated monitoring routine that detects and logs system errors.

Table 5.2: FaultDetection Tags (MainProgram)

Tag	Alias / Mapping	Type	Description
Current_State	—	DINT	Read: logged as SystemState in each fault record
FaultLog[0..4]	—	FaultLog[5]	Written: UDT array of fault records
FaultLogTime[0..4,0..6]	—	DINT[5,7]	Written: GSV wall clock timestamps per fault event

Tag	Alias / Mapping	Type	Description
Temp0Empty[0..4]	—	DINT[5]	Read: template to clear FaultLog on first scan (COP)
TempEmpty[0..4,0..7]	—	DINT[5,8]	Read: template to clear FaultLogTime on first scan (COP)
DoorFaultONS	—	BOOL	One-shot for DoorFault log entry
MotorFaultONS	—	BOOL	One-shot for MotorOnFault log entry
ReqFaultONS	—	BOOL	One-shot for PendReqFault log entry
SensorFaultONS	—	BOOL	One-shot for SensorFault log entry
EmergFaultONS	—	BOOL	One-shot for MovEmergFault log entry
MotorFaultBit	—	BOOL	Written: mirrors V_M0 for FaultProgram motor timer
OpenLSWFaultBit	—	BOOL	Written: mirrors V_DOLS for FaultProgram door timer
FloorLS1FaultBit	—	BOOL	Written: mirrors V_FLS1 for sensor fault detection
FloorLS2FaultBit	—	BOOL	Written: mirrors V_FLS2
FloorLS3FaultBit	—	BOOL	Written: mirrors V_FLS3
FloorLS4FaultBit	—	BOOL	Written: mirrors V_FLS4

Tag	Alias / Mapping	Type	Description
V_ILTS0	V_4O_Data.14	BOOL	Written: fault indicator – DoorFault active
V_ILTS1	V_4O_Data.13	BOOL	Written: fault indicator – MotorOnFault active
V_ILTS2	V_4O_Data.12	BOOL	Written: fault indicator – PendReqFault active
\FaultProgram.Faults[0].DoorFault	—	BOOL	Cross-program: door fault condition flag
\FaultProgram.Faults[0].MotorOnFault	—	BOOL	Cross-program: motor fault condition flag
\FaultProgram.Faults[0].PendReqFault	—	BOOL	Cross-program: pending request fault flag
\FaultProgram.Faults[0].SensorFault	—	BOOL	Cross-program: sensor conflict fault flag
\FaultProgram.Faults[0].MovEmergFault	—	BOOL	Cross-program: moving emergency fault flag
\FaultProgram.MajorFault	—	BOOL	Cross-program: major fault active flag

Table 5.2: Fault Code Summary

Fault Code	Fault Tag	Type	Detection Method	Condition
12	DoorFault	Minor	DoorFaultTimer.DN (TON 20 s)	Door open limit switch active continuously for more than 20 seconds

Fault Code	Fault Tag	Type	Detection Method	Condition
34	MotorOnFault	Minor	MotorFaultTimer.DN (TON 10 s)	Motor (V_M0) running continuously for more than 10 seconds
56	PendReqFault	Minor	Any TimerReq[N].DN (TON 15 s per floor)	Any floor request unfulfilled for more than 15 seconds
78	SensorFault	Minor	Combinatorial XIC/XIO rung	More than one floor limit switch active simultaneously
90	MovEmergFault	Major	State 100/400 with no floor LS active	Emergency stop triggered while elevator was between floors

Each fault is logged using a one-shot (ONS) to write the fault code, system state, major/minor classification, and a GSV wall clock timestamp into the corresponding FaultLog[] UDT entry. The FaultDetection routine in MainProgram reads these flags via cross-program tag references (\FaultProgram.Faults[]) and drives fault indicator lights V_ILTS0–V_ILTS2.

- Major Faults: Triggered by MovEmergFault (emergency while moving) or SensorFault. MajorFault blocks emergency reset until cleared.
- Minor Faults: MotorOnFault, DoorFault, and PendReqFault. Escalation to major fault occurs if the fault array fills with minor faults.

Table 5.3: FaultLog UDT Members

Member	Type	Description
FaultCode	DINT	Unique fault identifier (12, 34, 56, 78, or 90)
SystemState	DINT	Value of Current_State when the fault was logged
IsMajor	DINT	1 = major fault; 0 = minor fault

5.3 Part 9: Bonus Features

Subroutines: HMI_PentHouse and FlashingLight AOI. Adds a secure PIN-entry system for Floor 4 penthouse access via the HMI, and a reusable flashing light alert system as an Add-On Instruction.

Table 5.4: HMI_PentHouse Tags

Tag	Alias / Mapping	Type	Description
HMI_Pin_Input	— (Controller)	DINT	PIN value entered on the HMI numeric input field
HMI_Pin_Enter	— (Controller)	BOOL	Momentary bit driven by the HMI submit button
HMI_Pin_Correct	— (Controller)	BOOL	Set on correct PIN entry; drives HMI access-granted display
Request[4]	—	BOOL[32]	Written: Floor 4 request granted on correct PIN
RequestHMI[4]	—	BOOL[32]	Written/seal: HMI-originated Floor 4 request latch
FlashFloor4	—	BOOL	Written by AOI: 700 ms flash output for Floor 4 pending indicator
Current_Floor_Idle	—	DINT	Read: request blocked if elevator already at Floor 4
V_PB4	V_2I_Data.0	BOOL	In-car Pushbutton 4 – triggers PIN prompt
V_FPB4	V_2I_Data.8	BOOL	Floor Pushbutton 4 – triggers PIN prompt
V_IL4	V_4O_Data.0	BOOL	Written: Indicator Light 4 – driven by FlashFloor4
V_FIL4	V_4O_Data.8	BOOL	Written: Floor Indicator Light 4 – driven by FlashFloor4

Secure HMI Access – HMI_PentHouse

- Floor 4 requires the user to enter a PIN on the HMI. HMI_Pin_Input must equal 1234 for access to be granted.
- When a Floor 4 request is detected without a valid PIN, HMI_Pin_Required is set, prompting the PIN entry screen on the HMI.
- While waiting for PIN entry, the FlashingLight AOI drives FlashFloor4, causing V_IL4 and V_FIL4 to flash at 700 ms.
- On correct PIN entry (HMI_Pin_Enter high and HMI_Pin_Input = 1234), Request[4] is set and the elevator proceeds to Floor 4 normally via MultiCall and ThreeFloorElevator.

FlashingLight AOI

- A reusable AOI with two InOut parameters: In_Emergency (enable) and Light (output BOOL).

- Uses an internal 700 ms TON timer to toggle the Light output on and off while In_Emergency is high.
- Used in two contexts: EmergencyStopLight instance drives V_ILE5 during State 100/400; a second instance in HMI_PentHouse drives FlashFloor4 for the pending PIN indicator.

6 Conclusion

All nine parts of the Advanced Four-Floor Elevator project were completed successfully. The Method 3 state machine provided a clean framework for managing all seven operating states. Placing `StateMachine_Pre` and the emergency indicator rungs outside the MCR block ensured emergency transitions always fired correctly regardless of elevator operation state.

The `FloorStats` AOI made it straightforward to replicate per-floor statistics logic across all four floors without duplicating rungs. The `FlashingLight` AOI was reused in both the emergency indicator and the HMI Penthouse feature, demonstrating the value of well-designed, reusable AOIs.

The periodic fault detection task kept fault monitoring cleanly separated from elevator operation. The HMI Penthouse bonus feature combined controller-scope HMI tags, the `FlashingLight` AOI, and PIN-gated request logic into a self-contained routine that integrates naturally with the existing multi-call architecture.

Appendix Project Deliverables

Ladder Logic PDF

All subroutine PDFs were generated using print to PDF in Studio 5000, A3 landscape, power flow set to white.

Table A.1: Ladder Logic PDF Reference

PDF Filename	Program / Scope	Description
MainRoutine.pdf	MainTask : MainProgram	Master routine; first-scan init, subroutine call order, MCR block, emergency rungs
StateMachine_Pre.pdf	MainTask : MainProgram	Pre-MCR emergency transitions and START reset
StateMachinePose.pdf	MainTask : MainProgram	StateMachine_Post – floor arrival and door transitions, Next_State update
ThreeFloor.pdf	MainTask : MainProgram	Next_Floor resolution, Current_Floor_Idle tracking, V_M0 and V_CR0
InsidePanel.pdf	MainTask : MainProgram	In-car indicator lights IL1–IL3 sealed-in logic
Door.pdf	MainTask : MainProgram	Door solenoid, TimerA/B, ILO/ILC indicators, PBO extension logic
MultiCall.pdf	MainTask : MainProgram	Call/request latching, direction flags, RequestUp/Down derivation
StatAnal.pdf	MainTask : MainProgram	Per-floor statistics, timestamps, FloorStats AOI calls
HMI_Penthouse.pdf	MainTask : MainProgram	PIN-gated Floor 4 access, FlashingLight for pending indicator
EmergStop.pdf	MainTask : MainProgram	State 400 door-hold logic (JSR to Door subroutine)
FaultDetection.pdf	MainTask : MainProgram	Reads FaultProgram flags; logs FaultLog UDT; drives fault indicators
FaultMain.pdf	FaultTask : FaultProgram	Periodic fault timers and MovEmergFault combinatorial logic
FloorStatAOI.pdf	Add-On Instructions	FloorStats AOI logic
FloorStatTags.pdf	Add-On Instructions	FloorStats AOI tag listing

PDF Filename	Program / Scope	Description
FlashLigh.pdf	Add-On Instructions	FlashingLight AOI logic (700 ms toggle)
FlashLightTags.pdf	Add-On Instructions	FlashingLight AOI tag listing
ControllerTags.pdf	Controller Scope	HMI controller tags and I/O module references
MainTags.pdf	MainProgram Scope	Full main program tag listing
FaultTags.pdf	FaultProgram Scope	Fault program tag listing